

Rossmann Store Sales Dataset — Data Report

This is the public **Rossmann Store Sales** dataset (from Kaggle), used as the data foundation for **DemandLens** (RetailSense). Rossmann is a real European drug store chain with over 3,000 stores across 7 countries. The dataset comes from a Kaggle competition where the goal was to forecast daily sales for each store, 6 weeks into the future.

There are three files, and they work together as one system:

- **train.csv** — historical daily sales data. This is what a model learns from.
 - **test.csv** — future dates with sales hidden. This is what a model predicts for.
 - **store.csv** — background info about each store (type, competition, promotions). This adds context to both of the above.
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1. train.csv — Historical Sales

What it is: Daily sales records for 1,115 stores, covering about 2.5 years (Jan 2013 – Jul 2015). Over 1 million rows total — one row per store per day.

Why it exists: This is the training data — it contains the actual **Sales** numbers for each day, so a model can learn the relationship between a day's conditions (day of week, promo running, holiday, etc.) and how much the store sold.

Columns:

Column	Meaning
Store	Store ID number (links to <code>store.csv</code>)
DayOfWeek	1 = Monday ... 7 = Sunday
Date	Calendar date (YYYY-MM-DD)
Sales	Total revenue for that store on that day — this is the target variable to predict
Customers	Number of customers that day
Open	1 = store was open, 0 = closed
Promo	1 = store was running a promotion that day
StateHoliday	Marks a state holiday: a = public holiday, b = Easter, c = Christmas, 0 = none
SchoolHoliday	1 = that date overlapped with a public school closure

Sample rows:

Store	DayOfWeek	Date	Sales	Customers	Open	Promo
StateHoliday	SchoolHoliday					

1	5	2015-07-31	5263	555	1	1	0
1							
2	5	2015-07-31	6064	625	1	1	0
1							
3	5	2015-07-31	8314	821	1	1	0
1							
4	5	2015-07-31	13995	1498	1	1	0
1							

So on Friday, July 31, 2015, Store 1 had 555 customers and made €5,263 in sales, with a promo running and school holiday in effect that day.

2. test.csv — Prediction Targets

What it is: The same kind of daily records, but for a later 6-week window (Aug – Sep 2015), and **without** Sales or Customers — because those are exactly what needs to be predicted.

Why it exists: This simulates the real forecasting task — “given tomorrow’s conditions (day of week, promo, holiday), predict how much this store will sell.” It’s the file a trained model gets run against to produce forecasts.

Columns:

Column	Meaning
Id	Row identifier for this test set (used to match predictions back to rows)
Store	Store ID (links to store.csv)
DayOfWeek	1–7
Date	Calendar date to forecast
Open	1 = open, 0 = closed
Promo	1 = promo running that day
StateHoliday	Same coding as train.csv
SchoolHoliday	Same coding as train.csv

Sample rows:

Id	Store	DayOfWeek	Date	Open	Promo	StateHoliday	SchoolHoliday
1	1	4	2015-09-17	1	1	0	0
2	3	4	2015-09-17	1	1	0	0
3	7	4	2015-09-17	1	1	0	0
4	8	4	2015-09-17	1	1	0	0

This says: “On Thursday, Sept 17, 2015, Store 1 is open with a promo running — what will it sell?” That’s the question the model answers.

3. store.csv — Store Metadata

What it is: One row per store (1,115 rows) — static, non-changing background facts about each store. It doesn't have dates; it describes the store itself.

Why it exists: Two stores can have the exact same day-of-week/promo conditions and still sell very differently — because one is a bigger store, or has closer competition, or runs long-term promotions the other doesn't. This file supplies that context so the model can tell stores apart, not just days apart.

Columns:

Column	Meaning
Store	Store ID (the join key to train.csv / test.csv)
StoreType	Store format/model: a, b, c, or d (four differentiated store formats)
Assortment	Product range level: a = basic, b = extra, c = extended
CompetitionDistance	Distance in meters to the nearest competitor store
CompetitionOpenSinceMonth / CompetitionOpenSinceYear	When the nearest competitor opened
Promo2	1 = store participates in Promo2, a recurring/ongoing promotion program; 0 = it doesn't
Promo2SinceWeek / Promo2SinceYear	When the store joined Promo2
PromoInterval	Which months Promo2 relaunches in (e.g. Jan, Apr, Jul, Oct) — blank if not enrolled

Sample rows:

Store	StoreType	Assortment	CompetitionDistance	CompOpenMonth	CompOpenYear
		Promo2	Promo2Week	Promo2Year	PromoInterval
102	c	a	1270	9	2008
213	a	a	570	11	2007
314	a	a	14130	12	2006
405	c	c	620	9	2009
500	a	a	29910	4	2015
0	-	-	-	-	-

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store.csv  ┌──┐
            │  │ join on "Store" → gives every train/test row
train.csv  │  │                               full context: what day it was,
            │  │                               whether a promo ran, AND what
test.csv (no Sales) → same join, then the model predicts Sales
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In short: `train.csv` teaches a model what sales look like under given conditions, `store.csv` explains *why* stores differ from each other, and `test.csv` is the unseen future the model has to forecast for.